

NE523 Homework 1

Kyle Hansen

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Question 1 Two-species Boltzmann Equation

The Boltzmann equation can be derived by computing the transformation that a small element in phase space sees in a short time δt . From section 2, in the absence of collisions and considering only first-order δt terms, the particles at $(\mathbf{r}, \mathbf{v})$ will stream to $\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right)$ after a short time δt , or

$$f\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right) = f(\mathbf{r}, \mathbf{v}, t_0). \quad (1)$$

Note that δt must be small enough to neglect the higher-order terms. The effect of scattering can be denoted with $\left(\frac{\partial f}{\partial t}\right)_{coll}$, the net change in f over time due to scattering (into and out of $(\mathbf{r}, \mathbf{v})$). Then in time δt the total change in f is $\left(\frac{\partial f}{\partial t}\right)_{coll} \delta t$, and

$$f\left(\mathbf{r} + \mathbf{v}\delta t, \mathbf{v} + \frac{\mathbf{F}|_{t=t_0}}{m}\delta t, t_0 + \delta t\right) = f(\mathbf{r}, \mathbf{v}, t_0) + \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t. \quad (2)$$

Expanding the left-hand side as a Taylor series in t , this becomes

$$\cancel{f(\mathbf{r}, \mathbf{v}, t)} + (\nabla_r f \cdot \mathbf{v}) \delta t + \left(\nabla_v f \cdot \frac{\mathbf{F}}{m}\right) \delta t + \left(\frac{\partial f}{\partial t}\right) \delta t = \cancel{f(\mathbf{r}, \mathbf{v}, t)} + \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t \quad (3)$$

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{F}}{m} \nabla_v + \frac{\partial}{\partial t}\right) f(\mathbf{r}, \mathbf{v}, t) \delta t = \left(\frac{\partial f}{\partial t}\right)_{coll} \delta t \quad (4)$$

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{F}}{m} \nabla_v + \frac{\partial}{\partial t}\right) f(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial f}{\partial t}\right)_{coll}. \quad (5)$$

The same applies for particles of species g , so

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{F}}{m} \nabla_v + \frac{\partial}{\partial t} \right) f(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial f}{\partial t} \right)_{coll} \quad (6)$$

$$\left(\mathbf{v} \nabla_r + \frac{\mathbf{G}}{m} \nabla_v + \frac{\partial}{\partial t} \right) g(\mathbf{r}, \mathbf{v}, t) = \left(\frac{\partial g}{\partial t} \right)_{coll}. \quad (7)$$

The collision integral can be found by computing the rate of scattering into or out of $(\mathbf{r}, \mathbf{v})$ due to collisions of particles of both species.

Question 2 Differential Volume Invariant

Question 3 Maxwell-Boltzmann Moments

The distribution with arbitrary bulk velocity $\mathbf{v}_0$ is

$$f_0(\mathbf{v}) = n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{m|\mathbf{v} - \mathbf{v}_0|^2}{2kT} \right]. \quad (8)$$

Let $\mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} - \mathbf{v}_0$. Then distribution in the transformed coordinate is

$$f_0(\mathbf{w} + \mathbf{v}_0) = n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{m|\mathbf{w}|^2}{2kT} \right]. \quad (9)$$

Then the transformation disappears when integrating the zeroth moment, since the bounds of integration are infinite: $\int d^3w f_0(\mathbf{w} + \mathbf{v}_0) = \int d^3w f_0(\mathbf{w})$. Then the zeroth moment is

$$\int d^3w f_0(\mathbf{w}) = \int d^3w f_0(\mathbf{w} + \mathbf{v}_0) = \int d^3w n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{m|\mathbf{w}|^2}{2kT} \right]. \quad (10)$$

Transforming to spherical coordinates, the differential volume element becomes $d^3w = w^2 \sin \theta d\theta d\phi dw$, where $0 \leq \theta \leq \pi$ is the polar angle, $0 \leq \phi \leq 2\pi$ is the azimuthal angle, and $0 \leq (w = |\mathbf{w}|) < \infty$ is the speed, and the zeroth moment is

$$\int d^3w f_0(\mathbf{w}) = \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^\infty n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left[-\frac{mw^2}{2kT} \right] w^2 dw \quad (11a)$$

$$= 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty \exp \left[-\frac{mw^2}{2kT} \right] w^2 dw. \quad (11b)$$

With a change of variables $t = \frac{mw^2}{2kT}$, this becomes

$$\int d^3w f_0(\mathbf{w}) = 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty \frac{1}{2} \left(\frac{2kT}{m} \right)^{3/2} t^{1/2} \exp[-t] dt. \quad (12)$$

This integral can be evaluated using the gamma function $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$,

$$\int d^3w f_0(\mathbf{w}) = 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \frac{1}{2} \left(\frac{2kT}{m} \right)^{3/2} \Gamma\left(\frac{3}{2}\right) \quad (13)$$

$$= 4\pi n \left(\frac{m}{2\pi kT} \right)^{3/2} \frac{1}{2} \left(\frac{2kT}{m} \right)^{3/2} \frac{\sqrt{\pi}}{2}, \quad (14)$$

so the zeroth moment is

$$\boxed{\int d^3w f_0(\mathbf{w}) = n}. \quad (15)$$

The first moment is

$$\begin{aligned} \int d^3w (\mathbf{w} + \mathbf{v}_0) f_0(\mathbf{w} + \mathbf{v}_0) &= \int d^3w (\mathbf{w} + \mathbf{v}_0) f_0(\mathbf{w} + \mathbf{v}_0) \\ &= \int d^3w \mathbf{w} f_0(\mathbf{w} + \mathbf{v}_0) + \int d^3w \mathbf{v}_0 f_0(\mathbf{w} + \mathbf{v}_0). \end{aligned} \quad (16)$$

The second part of this integral can be evaluated easily using the zeroth moment, since $\mathbf{v}_0$ does not depend on $\mathbf{w}$; $\int d^3w \mathbf{v}_0 f_0(\mathbf{w} + \mathbf{v}_0) = \mathbf{v}_0 n$. The first part of this integral evaluates to zero, since the bracketed term in

$$\begin{aligned} \int d^3w \mathbf{w} f_0(\mathbf{w} + \mathbf{v}_0) \\ = n \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty \exp\left(-\frac{mw^2}{2kT}\right) w^3 dw \left[\int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \hat{w}(\theta, \phi) \right], \end{aligned} \quad (17)$$

where $\hat{w}(\theta, \phi)$ is the unit vector that points in the direction (θ, ϕ) , is equal to zero due to spherical symmetry.

Then the first moment is

$$\boxed{\int d^3v \mathbf{v} f_0(\mathbf{v}) = n \mathbf{v}_0}. \quad (18)$$

The second moment is

$$\int d^3v \left(\frac{1}{2} m v^2 \right) f_0(\mathbf{v}). \quad (19)$$

Applying the same transformation $\mathbf{w} = \mathbf{v} - \mathbf{v}_0$, this can be written

$$\int d^3v \left(\frac{1}{2}mv^2 \right) f_0(\mathbf{v}) = n \left(\frac{m}{2\pi kT} \right)^{3/2} \int d^3w \left(\frac{1}{2}m|\mathbf{w} + \mathbf{v}_0|^2 \right) \exp \left[\frac{-mw^2}{2kT} \right], \quad (20)$$

where

$$|\mathbf{w} + \mathbf{v}_0|^2 = (w_x + v_{0,x})^2 + (w_y + v_{0,y})^2 + (w_z + v_{0,z})^2 \quad (21a)$$

$$= (w_x^2 + 2w_x v_{0,x} + v_{0,x}^2) + (w_y^2 + 2w_y v_{0,y} + v_{0,y}^2) + (w_z^2 + 2w_z v_{0,z} + v_{0,z}^2) \quad (21b)$$

Then, translating to spherical coordinates, this becomes

$$\begin{aligned} |\mathbf{w} + \mathbf{v}_0|^2 = & (w_x^2 + 2 \sin \theta \cos \phi \cdot w v_{0,x} + v_{0,x}^2) \\ & + (w_y^2 + 2 \sin \theta \sin \phi \cdot w v_{0,y} + v_{0,y}^2) \\ & + (w_z^2 + 2 \cos \phi \cdot w v_{0,z} + v_{0,z}^2). \end{aligned} \quad (22)$$

Then, rewriting the integral in spherical coordinates yields

$$\begin{aligned} & n \left(\frac{m}{2\pi kT} \right)^{3/2} \left[\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^\infty dw \left(\frac{1}{2}mw^4 \right) \exp \left(\frac{-mw^2}{2kT} \right) \right. \\ & + \int_0^\pi d\theta \int_0^{2\pi} d\phi \int_0^\infty dw \left(m (\sin^2 \theta \cos \phi v_{0,x} + \sin^2 \theta \sin \phi v_{0,y} + \sin \theta \cos \phi v_{0,z}) w^3 \right) \exp \left(\frac{-mw^2}{2kT} \right) \\ & \left. + \frac{1}{2}mv_0^2 \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_0^\infty dw (w^2) \exp \left(\frac{-mw^2}{2kT} \right) \right]. \end{aligned} \quad (23)$$

The second line of the bracketed term evaluates to zero since $\int_0^{2\pi} \sin \phi d\phi = \int_0^{2\pi} \cos \phi d\phi = 0$. With the change of variables $t = \frac{mw^2}{2kT}$, this equation becomes

$$\begin{aligned} & n \left(\frac{m}{2\pi kT} \right)^{3/2} \cdot 4\pi \left[\int_0^\infty \frac{1}{2}m \left(\frac{2kTt}{m} \right)^2 \exp(-t) \cdot \frac{1}{2} \left(\frac{2kT}{m} \right)^{1/2} t^{1/2} dt \right. \\ & \left. + \frac{1}{2}mv_0^2 \int_0^\infty \frac{1}{2}m \left(\frac{2kTt}{m} \right) \exp(-t) \cdot \frac{1}{2} \left(\frac{2kT}{m} \right)^{1/2} t^{1/2} dt \right] \end{aligned} \quad (24a)$$